Protocol Design and Game Theoretic Solutions for Device-to-Device Radio Resource Allocation

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***Abstract*—Device to device (D2D) communication has been proposed to improve the resource efficiency and lighten the heavy load of the base station (BS) in LTE-Advanced systems. In a D2D-enabled LTE-A system, the resource efficiency is primarily determined by D2D/cellular mode selection and resource allocation. However, the D2D channel quality, the key factor for the BS to allocate resource, cannot be learned directly by the BS owing to the peculiarity of D2D communication. Rational user equipment (UEs) will take the advantage of the peculiarity to report their experienced D2D quality untruthfully for their selfish interests and consequently degrade the system efficiency. This so-called Unknown Channel Quality (UCQ) problem imposes a fatal impact on the resource efficiency and will limit the practicality of D2D communication. To overcome the UCQ problem, we propose to use game theory to analyze the peculiarity of D2D communication. In this paper, two practical D2D resource allocating protocols were investigated and the system efficiencies were analyzed to show the potential performance degradation when UCQ problem is not addressed. To circumvent the performance degradation caused by UCQ problem, a contract-based mechanism and the corresponding algorithms were proposed to eliminate the UE's incentive of reporting untruthfully. Numerical and simulation results validated the feasibility and the effectiveness of our approach.**

***Keywords*—Device-to-Device Communication, D2D, Resource Allocation, Game Theory, LTE-Advanced**

I. INTRODUCTION

LTE-Advanced is one of the most promising commercialized wide-area wireless network standard for achieving 4G requirements. Owing to the explosive growth of communications in cellular system, researchers endeavor to lighten the load of network traffic. One of the approaches schemed by the researchers for LTE-Advanced Release 12 standard is the Device-to-Device (D2D) communication.

D2D communication enables direct links between devices in proximity using the cellular spectrum without passing through the base station (BS). There are two main advantages for utilizing D2D communication. First, the devices can take advantage of the shorter distance to enjoy a higher throughput in a low-power transmission. Second, the BS can lighten its own load and serve more users if low transmit power between devices enables spectrum reusing. Radio resource allocation is needed in D2D network. Throughput fairness is a key metric of resource allocation in wireless network. In conventional cellular system, channel quality feedback is applied for link adaptation and resource allocation. As the channel quality affects the allocated throughput, both quantity of allocated radio resource blocks and the channel quality need to be considered to improve allocation efficiency and/or fairness. In conventional cellular systems, a BS could measure the channel quality itself and allocate radio resource accordingly. In D2D systems, in contrast, the channel quality can only be measured by the D2D communication pair. Hence, the BS allocates resource according to the information reported by the D2D user equipment (UE) in D2D systems. In such a case, Unknown Channel Quality (UCQ) problem may occur. In UCQ problem, a BS allocates radio resource according to the information provided by D2D UEs, but D2D UEs may report the information untruthfully for their selfish interests.

UCQ problem occurs when UEs are more aware of their own utilities instead of the overall system performance. To maximize the utility of an UE, the UE may report fake channel quality as long as such actions benefit itself. Such a selfish behavior misleads the BS at resource allocation and eventually makes the system inefficient. For example, if a UE cheats by telling the BS that it needs more resource to satisfy the D2D transmission demand owing to its poor channel quality, it will mislead the decision of BS to its favor, such as crowding out the resource of other honest UEs. Such selfish action leads to an unfair and inefficient operating point of the system. In order to formulate efficient systems in which both D2D and cellular modes are involved, we need to address the UCQ problem by considering the selfish behaviors of UEs using game theory.

Game theory has been popularly applied to model and analyze the hidden information problems [1], [2]. In our work, Stackelberg games are applied to formulate D2D resource allocation protocols into multi-stage game models. To solve the UCQ problem, we investigate how selfish UEs affect the network and cause inefficiency of the system under specific D2D resource requesting/allocation scenarios. Based on the analysis, contract theory [3] is viewed as a promising approach to solve the UCQ problem.

Contract is a popular solution to many game theory models involving incentive issues such as truth-telling or moral hazard.

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based on the users' private information. In such a system, users may report information untruthfully in order to retrieve unfair advantages. Furthermore, incorrect information will also lead the system to an inefficient equilibrium. Contract-based mechanisms can eliminate such outcomes by various approaches such as restricting players' strategy profiles or differentiating service properly. In this paper, we apply the contract theory to solve D2D resource allocation problem, specifically the UCQ problem. The proposed contract-based mechanism eliminates the incentive of the D2D UEs to report fake channel quality and achieves an efficient resource allocation maximizing the profit of the BS.

A. Our Contributions

We made the following contributions in this paper:

1) To best of our knowledge, we are the first to investigate the UCQ problem in D2D communication. We proved the existence of UCQ problem and showed that the system performance will significantly degrade since UEs report the fake channel quality. Therefore, a contract-based mechanism guaranteeing that all UEs report their quality honestly was proposed to allocate resource almost as efficiency as an omniscient model. With our mechanism, the UCQ problem is no longer as a cataclysm of the system in D2D communication.
2) Two LTE-D2D resource allocation protocol design candidates, UE-centric and BS-centric protocol design, are proposed in this paper for different application scenarios. We design game-theoretic mechanisms to allocate channel resource efficiently while addressing rational responses of UEs under four practical scenarios: Perfect Information UE-centric (PIUE), Unknown Channel Quality UE-centric (UQUE), Perfect Information BS-centric (PIBS) and Unknown Channel Quality BS-centric (UQBS). Through comparing the performance of different scenarios, we illustrate how UCQ problem may harm the efficiency of resource allocation in LTE-D2D systems and the effectiveness of the proposed contract-based mechanism.
3) We propose both exact and approximate algorithms to find the optimal contract sets which the BS may apply to maximize its profit while guaranteeing Nash equilibrium with truthful report. The simulation results show that the contract enhance the system performance in terms of efficiency in resource allocation and profit of the BS.

B. Outline

First, related works are discussed in Section II. Second, the D2D communication scenarios in LTE-Advanced systems are introduced in Section III. Two protocol designs and their multi-stage Stackelberg game models are also presented in Section III. Third, analysis of UE-centric and BS-centric protocol designs is presented in Section IV, V, respectively. Besides, a contract-based UCQ solution for BS-centric protocol design is in Section V. After the theoretical study, numerical and simulation results validating the proposed solutions are shown in Section VI. Finally, further discussions are given in Section

II. Related Works

D2D communication has been proposed in the literature as a practical and efficient solution in a cellular system by Fodor *et al.* [4]. Under LTE-Advanced networks, Doppler *et al.* introduced two different ways to establish D2D connections [5]. Yang *et al.* focused on peer discovery for D2D communication and proposed a distributed discovery protocol in LTE-A networks [6]. Furthermore, Grobmann, Kim *et al.*, and Golrezaei *et al.* proposed architectures and applications for D2D communication on video streaming [7], [8], [9].

Interference control is an important issue in D2D communication. Janis *et al.* analyzed the D2D system by concerning the local interference between cellular and D2D users [10]. Then, the uplink interference of D2D was analyzed by modeling as a Rayleigh fading channel by Kang and Shin [11]. Moreover, Pantisano *et al.* alleviated the uplink interference with a relay scheme between macrocell and femtocell [12]. Furthermore, Yue *et al.* and Zhang *et al.* used D2D interference to against eavesdroppers to improve the secrecy capacity [13], [14].

In order to realize D2D communication under the existing cellular system, spectrum leasing mechanism is an important issue. Wang *et al.* proposed an algorithm to allocate resource efficiently by reusing the resource [15]. Zhu *et al.* also proposed a QoS-based scheme for resource allocation by allowing the D2D user to share the spectrum with a cellular user [16]. When Chen *et al.* took the interference issue into consideration, they decided to only allocate downlink resource [17]. In a multicell system, Belleschi *et al.* introduced a centralized mathematical framework for resource allocation and mode selection [18].

For resource allocation problem, Game theory has become a new and effective approach. Wang *et al.* introduced Stackelberg game to deal with the resource sharing problem in cognitive radio networks [19]. Then, Yi *et al.* proposed a three-stages Stackelberg game in femto cell system to solve the resource allocation problem in a distributed way [20]. Pei and Liang presented a resource allocation protocol overlaying two-way cellular networks with the concept of Pareto Boundary [21].

Resource allocation in D2D communication with game-theoretic approaches has been proposed and most of popular methods are classified in [22]. Among these methods, auction is one of the most popular approach for resource allocation problems. For instance, Wang *et al.* designed the auction rules to achieve the system power efficiency while maintaining good channel quality [23], [24]. Considering the interference and spectrum efficiency, Xu *et al.* proposed a reversed iterative-combinatorial auctions (I-CAs) algorithm for spectrum allocation and reduced the interference between D2D and cellular users [25]. By utilizing the interference to guarantee the secrecy capacity, Yue *et al.* proposed an auction algorithm to tackle channel assignment problem [26]. Furthermore, Xu *et al.* used sequential second price auction rules to realize the resource allocation problem in D2D communication for preventing the users from high level interference [27]. In addition, some other approaches have been applied and studied

both formulated a two-stage Stackelberg game and made the system power efficient while guaranteeing the fairness between cellular and D2D users [28], [29]. Ye *et al.* also formulated a two-stage Stcakelberg game but to achieve a decentralized interference management through an iterative pricing scheme [30]. A dynamic mode selection algorithm for potential D2D users were presented and its solution, which is an evolutionary stable strategy was derived [31]. On the other hand, Zhang *et al.* modeled the system in which both D2D pairs and cellular users reach for a set of resource blocks. They made the users cooperate with each other in a coalition-game approach [32]. Li *et al.* worked for a similar goal but restricted the resource held in each cellular user before the coalition of D2D and cellular users, which implies the BS plays no roles [33]. Besides, Akkarajitsakul *et al.* allocated diagonal channels to cellular users and modeled the cooperative behaviors of D2D pairs in mode selection by a distributed coalition formation algorithm [34]. In order to optimize the system spectrum efficiency, Yu *et al.* proposed three different spectrum sharing modes with game approach in the system which contains a D2D pair and a cellular user in single cell [35]. Recently, an intercell D2D resource allocation problem has been studied [36] and a dynamic resource allocation scheme with service time prediction has been addressed [37].

As mentioned above, there exist multiple works studying the resource allocation problem in D2D communication overlaying cellular system in game-theoretic approach such as [28]–[31], [35] whose conducted with Stackelberg game, [32]–[34] with coalition game and [23]–[27] with auctions. Most of them made efforts on maximizing spectrum efficiency or power efficiency in systems that D2D users share the spectrum with cellular users. Such optimization relies on an assumption that all users will report their information, such as channel quality, truthfully. This assumption may be invalid when it comes to D2D communication, where the transmission occurs between devices instead of BS-device. In our previous work [38], we first explore the resource requesting protocol in D2D communication under a specialized BS-centric system scheme. We emphasize the challenges of UCQ problem and proposed a game-theoretic solution to the specific system. In this paper, we formally present two protocol design candidates named UE-centric protocol design and BS-centric protocol design for general LTE-D2D systems. Then we analyze how the information state impacts the UE's behavior and system performance. Eventually, the suggested protocol design and its contract-based solution of UCQ problem are proposed. An LTE-D2D simulator following the 3GPP standards for better evaluating the performance of the proposed solution is also implemented.

III. System and Game Model

In this section, the LTE-Advanced D2D communication scenarios we considered are introduced. Two protocol design candidates are presented and formulated into mathematical

TABLE I: Notation Table

Notation	Definition
β	Satisfactory weighting factor
c	Cost parameter
Q	Channel quality
B	Number of resource blocks
P	Price of per resource block
L	Current Load of BS
W	Bandwidth per resource block
SINR	Signal to interference plus noise ratio
TRU	Throughput
$U(B,Q,P)$	Utility function of UE
$R(B,Q,C)$	Profit function of BS
$U_b(B,Q)$	Benefit function of UE
$U_c(B,P)$	Cost function of UE
$R_r(B,P)$	Revenue function of BS
$C(B,L)$	Cost function of BS
X_{D2D}	X in D2D mode
X_{cell}	X in cellular mode
X^*	optimal choice of X

A. System Model and Assumptions

Prior to formulating our problem, we have to illustrate the system model and introduce our assumptions.

1) Operation of cellular and D2D services: An LTE-Advanced system offering two types of services: typical cellular service and D2D service is considered in this work. The D2D communication utilizes licensed spectrum and connected with the assistance of network infrastructures [5]. In the presented network, there is a BS, a spectrum license holder, and a set of UEs within its service area. The UEs have to request a permission from the BS in order to transmit through licensed wireless channel, either through cellular or D2D services. In D2D service, the resource blocks are allocated with a specific payment, which is determined by the channel quality of the D2D pair and the current load of the BS. In order to simplify the analysis, we assume that the uplink and downlink channels are symmetric in both cellular and D2D communications.

2) Selfish behavior and immediate utility: We assume the BS and UEs are rational and selfish. Therefore, they make decisions to maximize their immediate utilities. The utilities are represented through utility functions detailed in Section III-E.

3) Interference and power control: In order to serve both the cellular and D2D UEs in cellular networks, the power control mechanism is necessary for D2D communication to prevent the serious interference caused by D2D UEs in neighboring cells. In our model, the BS allocates uplink resource to D2D UEs and let the D2D UEs adopt the power control mechanism the same as the uplink power control mechanism in LTE-A systems [39], [40], where the transmit power increases proportional to the number of allocated resource blocks unless the maximum transmit power is reached.

4) Expected total throughput: The performance of the system in our work is measured by the expected total throughput. The throughput can be represented by the service quality, which depends on the channel quality. Given the average

and allocated resource blocks B, the consequent throughput can be conducted. Here we consider the channel quality Q as the average throughput per resource block. We apply, but not limited to, Shannon Capacity to describe the average throughput per resource block.

$$Q = W \times log_2(1 + \text{SINR}) \tag{1}$$

The expected total throughput is as follows:

$$\text{TRU} = B \times Q \tag{2}$$

5) BS's knowledge of the channel quality distribution of D2D pairs: We assume that all D2D UEs share the same utility functions but experience different channel qualities. Moreover, the BS has the a-priori knowledge on the distribution of the channel qualities of UEs. Note that exact channel quality of any specific UE is still unknown to the BS.

B. Problem Formulation

Our objective is to determine the optimal resource allocation of D2D service to maximize the profit of the BS. The BS is the one who holds the spectrum license. It not only determines the price/resource parameter to optimize efficiency and profit, but also has to guarantee that any rational and selfish UE will choose D2D service when this choice is more profitable to BS. A significant difference between the cellular and D2D services is the channel quality experienced by UEs. We assume that the cellular service has a fixed and predetermined price and resource quantity rule because the conventional cellular service has operated for several years and it is relatively difficult to alter the cellular payment rule. Based on the assumption that both UE and BS only consider immediate utilities in III-A2, the mathematical form of a BS's objective function is as follows:

$$\arg\max_{P_{D2D}} R(P_{D2D}, B_{D2D}, P_{cell}, B_{cell}, L), \quad \text{where } U_{D2D} \geq U_{cell} \tag{3}$$

The current load L represents the impact of resource allocated to previous D2D pairs and the current cellular service UEs, which is an exogenous variable for the BS. The goal of the BS is to maximize its profit through optimizing the profit from D2D service based on the current load and the pre-determined cellular contract.

In cellular mode, the BS can get the actual channel quality itself. In D2D mode, due to the property of D2D communication, the BS can hardly do a prior or a posterior validation on the reported channel quality owing to the signaling and processing overhead in reality [4]. The main reason is that the data transmitted between UEs without passing through the BS. In order to get more resource or cheaper price, rational UEs may report fake channel quality for their own interest. This is so called UCQ problem. The main goal of this work is to properly analyze the rational behavior of BS and UEs

C. Protocol Design

In order to provide D2D service in addition to cellular service under LTE-Advanced systems, two protocol design candidates: UE-Centric protocol design and BS-Centric protocol design are presented. The difference between these two protocols is the degree of freedom of UE. The following paragraph will show the signaling flow of these two protocols.

1) UE-Centric Protocol Design: In UE-centric protocol design, the UE has the right to determine the allocated resource blocks in D2D communication. This protocol design is preferred and has been suggested by some auction-based mechanisms [23]-[27] for D2D communication because the quantity controlled by the demand side (UE) are usually more efficient. The signaling flow of UE-centric protocol design works as follows.

(1) D2D communication protocol is initiated by UE. The requesting UE sends a signaling message to BS to request for D2D communication with targeting UE.
(2) The BS identifies the targeting UE in request, and then sends an acknowledgement with the information of the targeting UE to help the requesting UE for peer discovery. The acknowledgement also includes the message of available resource blocks and prompts the requesting UE to start the D2D channel measurement.
(3) Upon receiving the acknowledgment, the requesting UE does peer discovery and then performs D2D channel quality estimation with the targeting UE.
(4) The requesting UE reports the information of channel quality to the BS.
(5) The BS computes the D2D price based on the reported channel quality and sends the per channel quantity price of D2D/cellular service P_{D2D}, P_{cell} to the requesting UE.
(6) The requesting UE chooses either cellular or D2D service and sends the response to the BS. If the UE chooses the D2D service, the response includes the amount of resource quantity B_{D2D} which the UE requests from the BS.
(7) The BS confirms the service mode and allocates the radio resource blocks to the UE for D2D communication.
(8) The UE starts to transmit data according to the mode it selects.

2) BS-Centric Protocol Design: In BS-centric protocol design, the BS has the right to determine both price and allocated resource quantity in D2D communication. The only right of the UE is to choose using D2D service or cellular service. The signaling flow is as follows:

(1)-(4) Same as UE-centric protocol design
(5) The BS computes D2D price and allocated resource quantity based on the reported channel quality and sends the D2D/cellular offer $[(P_{D2D}, B_{D2D}), (P_{cell}, B_{cell})]$ to the requesting UE.
(6) The requesting UE chooses cellular or D2D service and sends the response to the BS.

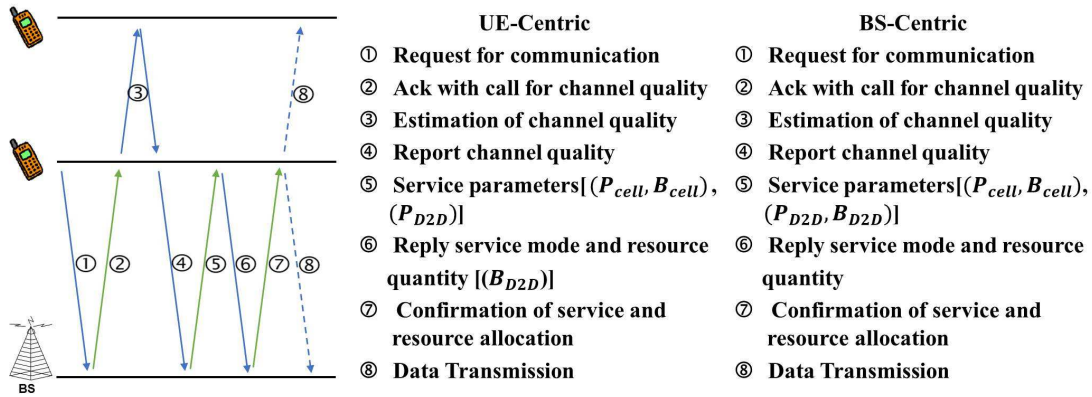

Fig. 1: UE-centric and BS-centric Protocol Design

D. Model Protocols as Multi-stage Stackelberg Games

The transmission requests and service mode selection, as illustrated in UE-centric and BS-centric protocol design, can be modeled as sequential games, that can be analyzed by game theory. Before formulating game models, some essential definitions of games should be introduced first.

Nash Equilibrium represents a rational and stable outcome of a game. It can also be considered as a reasonable prediction on the outcome of the system given the interactions between players. The formal definition of Nash Equilibrium is as follows:

Definition 1. *Let X_i denote the action set of player i, a Nash equilibrium is a strategy profile $S^* = \{x_1^*, x_2^*, ..., x_n^*\}$ which $\forall x_i' \neq x_i^* \in X_i$, $\mathrm{Util}_i(x_i^*, S_{-i}^*) \geq \mathrm{Util}_i(x_i', S_{-i}^*)$.*

where x_i^* is player i's Nash equilibrium strategy of equilibrium S^*; $\mathrm{Util}_i(x_i, S_{-i})$ is player i's utility when it chooses strategy x_i and other players choose a strategy profile S_{-i}.

For games containing multiple steps such as sequential games, the Nash equilibrium needs some refinements. A sequential game can be viewed as a composition of multiple subgames by forming the game as a game tree. Before refining Nash equilibrium, the definition of game tree and subgame should be introduced first.

Definition 2. *A game tree is a directed graph whose nodes are positions in a game and whose edges are moves.*

Definition 3. *A subgame G_y with root node y is a branch of a game tree G_x with root node x .*

Subgame perfect Nash equilibrium (SPNE), usually analyzed by backward induction, is a popular Nash equilibrium refinement in sequential games. The definition of SPNE is as follows.

Definition 4. *A subgame perfect Nash equilibrium of a sequential game G is a strategy profile $S^* \in S$ such that, for each subgame G_x of G, S_x^* is a Nash equilibrium of G_x.*

where x is a node in game G, G_x is a game tree whose root node is x, and S_x^* is Nash equilibrium strategy profile in

Since the definitions are defined, we can transform the requests in protocol designs to Stackelberg games. A Stackelberg game is a sequential game which consists of a leader and a follower, each performs actions in a predefined sequential order. In a conventional Stackelberg game, the leader moves first and then the follower moves next. The leader knows ex-ante that the follower observes its action; and chooses the best strategy for itself after surveying each final outcome of the follower's best response. The Stackelberg game models of UE-centric protocol design and BS-centric protocol design are listed below:

1) UE-centric Resource Request Model: The UE-centric protocol design can be modeled as a three-stage Stackelberg game. First, the UE sends a transmission request to the BS. The message includes the channel quality and other necessary information. The UE acts as a leader. Second, the BS, playing the role of follower, responds by offering resource price of D2D service and contract (resource price and quantity pair) of cellular service. Third, the UE reports the desired transmission mode and resource quantity based on the offered contract. If the UE chooses D2D mode, it will determine the amount of required resource that maximize its utility.

2) BS-centric Resource Allocation Model: The BS-centric protocol design can also be simplified to a three-stage Stackelberg game. The first stage is the same as UE-centric protocol design. However, in the second stage, the BS responds with both the resource price and quantity in D2D/cellular service. Therefore, in the third stage, the UE can only do mode selection based on the offered contracts and have no freedom to determine the resource quantity.

E. Game Settings

As mentioned in previous subsection, both UE-centric protocol design and BS-centric protocol design can be modeled as three-stage Stackelberg games with some minor difference in game flows. Now, we define basic elements of proposed three-stage Stackelberg game models.

1) **Players and Actions**: This game has two players: the BS and the UE. The actions of the BS are to determine the cellular service contract and D2D price/contract in

UE's actions are to determine the information in the request message, which service mode to transmit, and the allocated resource quantity if in UE-centric model.

2) **Utility function of UEs**: The objective of the UE is to maximize its utility. The utility function has two components:
   a) Benefit $U_b(Q, B)$: The UE benefits from the increasing of expected total throughput $\mathrm{TRU} = B \times Q$. We assume U_b is a strictly concave increasing function to $B \times Q$. Furthermore, marginal benefit $\frac{\partial U_b(Q,B)}{\partial B}$ increases in quality Q.
   b) Payment $U_c(B, P)$: The payment of UE is the product of resource quantity B and the per resource price P.

   Given these two components, the utility function is as follows:

$$U(Q, B, P) = U_b(Q, B) - U_c(B, P) \tag{4}$$

3) **Profit function of the BS**: The objective of the BS is to maximize its profit, therefore its utility function is the profit it earns. The profit of the BS is the revenue it receives minus the cost of offering the services. We define the profit function with two components:
   a) Revenue $R_r(B, P)$: The revenue of the BS is the payment of the UE. $R_r(B, P) = U_c(B, P) = B \times P$.
   b) Cost $C(B, L)$: Owing to the scarcity of the resource blocks, we assume the cost function is a convex increasing function of resource blocks B. Meanwhile, we assume the cost function is an increasing function of current load L, where $\frac{\partial C(B,L)}{\partial L} > 0$. The current load L is an exogenous variable for the BS in our model. A higher load L suggests few available resource and therefore higher cost if allocated to the UE. Hence the profit function of the BS can be represented as follows:

$$R(B, P, L) = R_r(B, P) - C(B, L) \tag{5}$$

4) **Four Different Scenarios**: In order to deal with UCQ problem, we consider two information states in both UE/BS-centric protocol designs. Table II displays the relationships among these four scenarios.
5) **Game Process**: First, a communication request containing the D2D channel quality Q is sent from an UE. Second, the BS determines the offers of services: D2D price (and also the quantity if in BS-centric model) and cellular contract to the UE. Last, the UE chooses the best response (service mode selection and the request of D2D resource quantity if in UE-centric model) based on its own utility function and offered contracts. The game process is illustrated in Table III.

IV. UE-centric Resource Request Model

In this section, we first analyze the scenarios of UE-centric protocol design without the UCQ problem. Then, the UCQ problem is taken into consideration to understand its

TABLE II: Four Different Scenarios

Information State	Scheme	
	UE-centric	BS-centric
Perfect Information	PIUE Model	PIBS Model
UCQ State	UQUE Model	UQBS Model

TABLE III: Game Process

Stage 1	UE sends channel quality Q to the BS
Stage 2	BS sends fixed cellular contract and D2D contract (P_{D2D}) or $(P_{D2D},\ B_{D2D})$
Stage 3	UE selects service mode and determines D2D resource B_{D2D} (if necessary)

A. Perfect Information Scenario

The Perfect Information UE-centric resource request (PIUE) scenario represents the ideal case that the UEs always truthfully report their channel quality and therefore the BS has perfect information. It helps us to grasp insights into the behavior and reaction of BS and UEs in the proposed D2D system.

According to Section III, the action for the UE is to determine the allocated resource blocks B_{D2D} in D2D service. If the UE decides to transmit in cellular service, then its strategy would be $B_{D2D} = 0$. Otherwise, it would be $B_{D2D} = B^*_{D2D}$. On the other hand, action for the BS is $[(P_{cell}, B_{cell}), (P_{D2D})]$,which is the tuple of two service contracts. Here we assume that P_{cell} and B_{cell} are fixed and predetermined. This reflects the fact in real world that the cellular service contract is signed by the users in advance.

In perfect information state, the three-stage Stackelberg in Section III can be simplified to a two-stage Stackelberg game because the only strategy for UE in first stage is to report the experienced channel quality honestly. Then, we derive the SPNE through backward induction. The first step is to derive the best response of UEs given the contracts announced by the BS. The UE will report the optimal quantity of resource for itself that maximizes its utility, given the specific price. That is, $B^*_{D2D} = \arg\max_B U_{D2D}$. The optimal quantity of resource can be written as a function of given price. Since the $U_b(\cdot)$ is a strictly concave increasing function to B and $U_c(\cdot)$ is linear to B under a specific $\bar{P}$; the optimal quantity and the corresponding utility of UE are as follows:

$$B^*_{D2D}(Q) = \{B | \frac{\partial U_b(Q, B)}{\partial B} = \bar{P}\} \tag{6}$$

$$U^*_{D2D}(Q) = U_b(Q, B^*_{D2D}) - U_c(B^*_{D2D}, \bar{P}) \tag{7}$$

Next, the BS will maximize its profit based on its prediction of the UE's response to its action. The BS predicts the amount of resource blocks that the UE will purchase under every given price $\bar{P}$, then determines the price per resource block to maximize its profit, that is, $P^*_{D2D} = \arg\max_P R(B, P, L)$. As above, we can get the price for resource by applying $\frac{\partial R(P)}{\partial P} = 0$. The consequent equation is

$$\frac{\partial R(P)}{\partial P} = \frac{\partial R_r(P)}{\partial P} - \frac{\partial C(P)}{\partial P}$$

$$\partial R_r(B^*_{D2D}(P), P) \quad \partial C(B^*_{D2D}(P), L)$$

Finally, the optimal price P^*_{D2D} can be derived in (8) by substituting B^*_{D2D} with P according to (6). Then, the optimal number of resource blocks B^*_{D2D} can be conducted from (6).

B. Criteria of Mode Selection in PIUE Scenario

After conducting the optimal price and allocated resource blocks of D2D mode in PIUE scenario, we have to give several criteria of selecting D2D mode or not. According to the backward induction, the conducted P^*_{D2D}, B^*_{D2D} must be the SPNE of PIUE scenario if there is no cellular mode service. Taking the cellular mode into consideration, we bring out a lemma to derive the properties of SPNE in PIUE scenario. Notice that the proofs of all lemmas and theorems, if not stated otherwise, are provided in Appendix.

Lemma 1. *The D2D service is an SPNE for the UE and the BS if and only if the strategy satisfies the following inequalities:*

$$B^*_{D2D} > 0 \quad ; \tag{9}$$

$$U^*_{D2D}(Q, B^*_{D2D}, P^*_{D2D}) \geq U_{cell}; \tag{10}$$

$$R^*_{D2D}(B^*_{D2D}, P^*_{D2D}, L) \geq R_{cell}, \tag{11}$$

where

$$\begin{aligned} 0 &= \left.\frac{\partial U^*_{D2D}(Q, B_{D2D}, P^*_{D2D})}{\partial B_{D2D}}\right|_{B^*_{D2D}} \\ &= \left.\frac{\partial R^*_{D2D}(B^*_{D2D}, P_{D2D}, L)}{\partial P_{D2D}}\right|_{P^*_{D2D}}. \end{aligned}$$

When (9)-(11) hold, the necessary condition of Lemma 1 shows that choosing D2D communication is rational for the UE in PIUE scenario, and the optimal D2D price and amount of allocated resource blocks can be derived accordingly. However, the relationship between channel quality Q and optimal amount of resource blocks B^*_{D2D} is still unclear. Hence, we discuss a special case that the optimal amount of resource blocks requested by the UE increases with respect to channel quality in Lemma 2.

Lemma 2. *If benefit function of UE is a strictly concave increasing polynomial function to the expected total throughput* TRU*, the optimal amount of resource blocks requested by the UE increases with respect to channel quality.*

C. Unknown Channel Quality Scenario

In PIUE scenario, we assume that UEs always truthfully report their experienced channel quality in D2D service. However, the BS need to face the UCQ problem in reality. In order to analyze the impact of UCQ problem, we investigate the Unknown Channel Quality UE-centric resource request (UQUE) scenario to understand its characteristic.

In UQUE scenario, the UE not only chooses a strategy for B_{D2D} but also determines the channel quality Q reported to the BS. It can mislead the BS with a better/worse channel quality if the fake report could benefit itself. In such a scenario, the BS must take the possibility of a lying UE into consideration in order to avoid inefficient outcome. The following Theorem 1 will show the rationality of the fake report of UE.

Theorem 1. *If selecting D2D service is a Pareto improvement of the system and the equilibrium D2D price varies under different Q, honest strategy will never be a Nash Equilibrium.*

Theorem 1 indicates that the honest strategy is not a Nash equilibrium in UQUE game. Theorem 1 can be proved by first finding the truth-telling condition, then proving that the condition contradicts the assumption in our system model. Now, we propose the following lemma to indicate the existence of SPNE in UQUE scenario.

Lemma 3. *$\forall Q \in R$, the only SPNE existing in D2D mode of UQUE game satisfies the equation:*

$$\lim_{n\to\infty} U_{D2D}(Q_{lie,n}) + R_{D2D}(Q_{lie,n}) = U_{cell} + R_{cell} \tag{12}$$

Proof. Theorem 1 shows that the selfish UE will lie to the BS by replying a lower/higher channel quality for a lower price. Define $Q_{lie,k}$ to represent the channel quality the UE will reply to the BS after $k-1$ round best response analysis between the BS and UE. At initial state, the UEs will choose a $Q_{lie,1}$ to reply based on the analysis of the relationship between Q_{D2D} and $U_{D2D}(Q, B, P)$ according to (29).

According to Definition 4, once the BS awares that UEs would lie to it by reporting $Q_{lie,1}$, it must take actions otherwise the outcome will not be an SPNE. The BS, in order to maximize its own profit, tries to figure out the real channel quality from the $Q_{lie,1}$. In our system model, the BS knows the utility function of UEs, it can fully derive the relationship between $Q_{lie,1}$ and the real Q_{D2D} if there is no constraint on delay or computing resource. Once the BS derived the real Q_{D2D}, it can choose the best response for itself by setting a price $P_{lie,1}$ and diminish the profit lose from the selfish UEs.

After that, the UE will notice it and compute a new fake channel quality $Q_{lie,2}$ based on the analysis of $P_{lie,1}$. Assume $Q_{lie,1} < Q_{D2D}$ without loss of generality, the $Q_{lie,2} < Q_{lie,1}$ always holds because the analysis follows the same policy. Then the BS notice it again, deriving the Q_{D2D} from $Q_{lie,2}$ and set a new price $P_{lie,2}$ as a response. The process iterates repeatedly and we derive the following inequality:

$$Q_{lie,1} > Q_{lie,2} > ... > Q_{lie,k} > Q_{lie,k+1} > ..., \quad \text{if } Q_{lie,1} < Q_{D2D} \tag{13}$$

In k^{th} round, after the UE and BS analyze the strategy of each other, the $Q_{lie,k}$ goes lower than $Q_{lie,k}$. The iterating process will be terminated when the reported channel quality approximate to a breakeven point of applying D2D mode. It implies the following equation:

$$\lim_{n\to\infty} U_{D2D}(Q_{lie,n}) + R_{D2D}(Q_{lie,n}) = U_{cell} + R_{cell} \tag{14}$$

Therefore, the only SPNE exists in $U_{D2D}(Q_{lie}) + R_{D2D}(Q_{lie}) = U_{cell} + R_{cell}$. □

Notice that when the above SPNE is achieved, the BS has difficulty measuring the real channel quality of the D2D service. In fact, the BS knows nothing about the real channel condition from the B_{lie} and therefore can not allocate resource efficiently.

Lemma 3 shows that the UE never discloses its true channel

report a little bit higher than the breakeven point of offering D2D service regardless of its true channel quality. Such an SPNE suggests that there is no reliable information from reported channel quality. In order to prevent the loss when the cost of resource exceed the revenue owing to the plethoric resource request from UE, the BS may adopt some extremely conservative policies. A safe strategy for the BS is setting a relatively high price to obviate the scenario. However, the strategy may eliminate significant amount of D2D communication opportunities since UEs who potentially benefits from the D2D services may be excluded due to the high resource price. Therefore, the UE-centric protocol design is not an efficient solution for D2D communication under the existence of UCQ problem.

V. BS-centric Resource Allocation Model

In Section IV, we study the UCQ problem when the BS does not know the real channel quality between UEs. We conclude that the UE-centric protocol design is not a suitable protocol for D2D communication: the right for UEs to determine the resource quantity will give them an incentive to lie about their channel quality in order to get a better price. Consequently, an alternative approach is necessary.

The BS-centric protocol design now is a promising candidate: the BS determines both the price and quantity after UEs request for communication service, that is to say, the BS fully controls the resource allocation in both D2D and cellular services. We will show that through limiting the freedom of the UE, the UCQ problem can be eliminated efficiently.

A. Perfect Information Scenario

Perfect Information BS-centric resource allocation (PIBS) scenario can be simplified to a two-stage Stackelberg game similar to Section IV-A and analyzed through the backward induction.

1) UE's strategy: In the PIBS scenario, the UE can only choose to accept the D2D contract given by the BS to transmit in D2D mode, or reject it to transmit in cellular mode. By Lemma 1, the UE will reject all D2D contracts those make $U_{D2D} < U_{cell}$. In other words, the UE will accept any D2D contract as long as a contract satisfies the following inequality:

$$U_{D2D}(Q, B, P) = U_b(Q, B) - U_c(B, P) \geq U_{cell} \qquad (15)$$

2) BS's strategy: In order to maximize the profit, BS should determine the most profitable D2D resource price-quantity contract under the constraint of UE's willingness to accept the offer.

Taking the strategies of both UE and BS into consideration, we can conduct Lemma 4.

Lemma 4. *$U^*_{D2D} = U_{cell}$ holds when BS maximizes its profit in D2D service.*

Lemma 4 implies that the optimal price P_{D2D} for BS can be presented as a function of resource quantity, $P(B_{D2D})$, by solving the boundary condition of (15). We should derive the resource quantity which maximizes the profit: $B^*_{D2D} =$

Since $U_{D2D}(Q, B, P) = U_b(Q, B) - U_c(B, P) = U_{cell}$, we can simply adjust the payment price after the derivation when $U_{cell} \neq 0$. The equilibrium B and P is as follows:

$$B^*_{D2D} = \arg\max_{B_{D2D}} \{U_b(Q, B_{D2D}) - C(B_{D2D}, L)\} \qquad (16)$$

$$P^*_{D2D} = \frac{U_b(Q, B^*_{D2D}) - U_{cell}}{B^*_{D2D}} \qquad (17)$$

3) System Performance: Lemma 4 implies that the BS can always extract the UE's utility into profit by increasing the price, as long as it satisfies the willingness constraint (3). Under a discrete pricing scheme, the BS can achieve an SPNE in D2D system by slightly decreasing the price from its best strategy.

Observing the results in (16) and (17), we find out that as the channel quality increases, the price P_{D2D} which the BS offers will increase. However, the resource quantity that the BS allocates will slightly increase when the channel quality increases because of the concavity of $U_b(Q, B)$ and convexity of $C(B, L)$.

B. Unknown Channel Quality Scenario

According to the above analysis, we conclude that the BS-centric protocol design is efficient in perfect information state. We want to know if this protocol design is practical in the Unknown Channel Quality (UQBS) scenario. Given the a-priori knowledge on the distribution of the channel qualities, we assume that the BS can distinguish N different types of channel quality: $Q_1, Q_2, Q_3, \cdots, Q_N$. Before the analysis, we first define the expected overall profit of the BS.

Definition 5. *The expected overall profit of the BS is the sum of profit from all types of UEs it serves. $R^{TOT} = a_1R^1 + a_2R^2 + a_3R^3, \cdots, a_NR^N$, where a_k is the number of UEs of type k.*

1) Solution of UCQ Problem with Offered Contracts: In order to solve UCQ problem and maximize its own profit, the BS has to determine the offered contracts. We propose three algorithms to help the BS to find the feasible contract set, the optimal contract set, and the optimal serving types.

Feasible contract set is a set of contracts which are able to distinguish UEs from its neighboring channel quality types. The formal definition of feasible contract sets is as follows:

Definition 6. *Given channel qualities $Q_1, Q_2, \cdots, Q_N$, a set of contracts $\{(B_i, P_i) \mid i \in [1, N]\}$ is feasible if and only if $U_{D2D}(Q_i, B_i, P_i) \geq U_{D2D}(Q_i, B_{i-1}, P_{i-1})$ and $U_{D2D}(Q_i, B_{i-1}, P_{i-1}) \leq U_{D2D}(Q_{i-1}, B_{i-1}, P_{i-1})$, $\forall i \in [2, N]$.*

Theorem 2 illustrates an useful property of the feasible contract set.

Theorem 2. *A feasible contract set is a set of contracts guaranteeing that all UEs report channel quality honestly.*

Theorem 2 indicates that given any feasible contract set, UEs will report their experienced channel qualities truthfully in order to receive the designated contracts specifically for

of UEs if a valid feasible contract set is chosen. The following Algorithm 1 is proposed to find a feasible contract set.

Algorithm 1 Determine Feasible Contract Set

Input: Channel Quality of each type: $Q_i, i \in [1, N]$;
Output: Corresponding price and resource quantity of channel quality: $B_i, P_i i \in [1, N]$;
1: Solve $(B_1^{ideal}, P_1^{ideal})$ by equating $U_{D2D}(Q_1, B_1^{ideal}, P_1^{ideal}) = U_{cell}$.
2: Find the least integer (B_1, P_1) satisfying $U_{D2D}(Q_1, B_1, P_1) \geq U_{D2D}(Q_1, B_1^{ideal}, P_1^{ideal})$
3: **for** each $Q_i, i \in [2, N]$ **do**
4: Find an integer contract (B_i, P_i) satisfying $U_{D2D}(Q_i, B_i, P_i) \geq U_{D2D}(Q_i, B_{i-1}, P_{i-1})$ and $U_{D2D}(Q_{i-1}, B_i, P_i) \leq U_{D2D}(Q_{i-1}, B_{i-1}, P_{i-1})$
5: **end for**

The contract set conducted from Algorithm 1 is feasible, which implies that every UE will choose its designated contract. The computing complexity of the algorithm is linear to number of UE's type, $(O(N))$.

In addition to Algorithm 1, we propose Algorithm 2 to find the subset of feasible contracts sets which maximizes the profit of the BS. First, we formally define the optimal contract set.

Definition 7. *An optimal contract set is a feasible contract set which maximizes the profit of the BS corresponding to a specific channel quality set $Q_k \in Q$.*

Algorithm 2 Determine Optimal Contract Set

Input: Channel quality in given channel quality set: $Q_i \in Q$;
Output: Corresponding price and resource quantity of channel quality: B_i, P_i;
1: Sorting $Q_i \in Q$ to get Q_{K_i}, $i \in [1, \textit{size of } Q_i]$,
2: Solve $(B_{K_1}^{ideal}, P_{K_1}^{ideal})$ by equating $U_{D2D}(Q_{K_1}, B_{K_1}^{ideal}, P_{K_1}^{ideal}) = U_{cell}$.
3: Find the least integer contract (B_{K_1}, P_{K_1}) satisfying $U_{D2D}(Q_{K_1}, B_{K_1}, P_{K_1}) \geq U_{D2D}(Q_{K_1}, B_{K_1}^{ideal}, P_{K_1}^{ideal})$
4: **for** each $Q_{K_i}, i \in [2, \textit{size of } Q_i]$ **do**
5: Solve $(B_{K_i}^{ideal}, P_{K_i}^{ideal})$ which maximizes the profit of the BS with $U_{D2D}(Q_{K_i}, B_{K_i}^{ideal}, P_{K_i}^{ideal}) = U_{D2D}(Q_{K_{i-1}}, B_{K_{i-1}}, P_{K_{i-1}})$.
6: Find the least integer contract (B_{K_i}, P_{K_i}) satisfying $U_{D2D}(Q_{K_i}, B_{K_i}, P_{K_i}) \geq U_{D2D}(Q_{K_i}, B_{K_i}^{ideal}, P_{K_i}^{ideal})$
7: **end for**

Algorithm 2 conducts the set of optimal contract set for the BS, which is a subset of the feasible contracts. However, it has a higher complexity than Algorithm 1 because of the computing of contracts with maximum profit of BS. The complexity of Algorithm 2 is linear to the computation of numerical solutions in line 5, Algorithm 2.

Algorithm 2 outputs an intermediate solution for the optimal contract set as it assumes that the serving types of each set of UEs are already determined. The BS still has to determine the optimal serving types of UE set to maximize its overall profit.

Definition 8. *The optimal serving types are types of UE with of BS is maximized by offering optimal contract set to these types.*

A naive algorithm for finding the optimal serving types is given in Algorithm 3. The complexity of Algorithm 3 is $O(2^N)$ because the number of serving types set grows exponentially over number of types.

Algorithm 3 Determine Optimal Serving Types

Input: Channel Quality of each type: $Q_i, i \in [1, N]$;
Output: Optimal contract set $S_C^* = \{(B_k, P_k)\}$
$S_C^* = \{(B_k, P_k) \mid k \in [1, N]\}$
for each possible channel quality set Q **do**
Use Algorithm 2 to Derive optimal contract set S_C for corresponding Q
Calculate expected profit $R(S_C)$ for the BS
if $R(S_C) > R(S_C^*)$ **then**
$S_C^* = S_C$
end if
end for

2) A Special Case - Uniform Resource Allocation: Since the high complexity of Algorithm 3 sometimes makes it impractical in real world, we would like to proposed a low-complexity solution in some specific scenarios. We first introduce a set of utility functions of UE and profit functions of BS that satisfy all the properties in Section III.

$$U(Q, B, P) = \beta \ln(BQ) - PB \tag{18}$$

$$R(B, P) = PB - c \times L \times B^{\alpha} \qquad \alpha > 1 \tag{19}$$

These functions are linear approximate where the proper parameters β and c can be chosen by regression methods. With these specific functions, we observe some useful properties that make the optimization more efficient.

Lemma 5. *Each type of UEs which the BS serves will be allocated the same resource $\bar{B} = (\frac{\beta}{\alpha c L})^{1/\alpha}$ in equilibrium when the utility functions $U_{D2D}^k(Q_k, B_k, P_k) = \beta \ln(B_k Q_k) - P_k B_k$ and profit function $R(B, P)^{TOT} = \sum_{i=1}^{N} P_i B_i - c \times L \times B_i^{\alpha}$,.*

Proof. The same method can be used to solve N types of UEs. The following equations must hold when the profit of the BS is maximized.

$$\begin{aligned}
U_{D2D}^1 &= \beta \ln(B_1 Q_1) - P_1 B_1 = U_{cell} \\
U_{D2D}^2 &= \beta \ln(B_2 Q_2) - P_2 B_2 = U_{cell} + \beta \ln(\frac{Q_2}{Q_1}) \\
&\vdots \\
U_{D2D}^N &= \beta \ln(B_N Q_N) - P_N B_N = U_{cell} + \beta \ln(\frac{Q_N}{Q_1})
\end{aligned}$$

From the above equations, the constant term will disappear in the process of differentiation, so the optimum (B_k, Q_k) pair of the BS remains unaffected by other types of UEs. We can solve each B_k $\forall k \in [1, N]$ as follows:

The BS determines to serve some types of UEs, allocating resource $B_k = \bar{B} = (\frac{\beta}{\alpha c L})^{1/\alpha}$. In other words, all UEs which the BS serves will be allocated the same resource quantity. □

With the same resource quantity, these contracts must have the same resource price, which means that the BS offers the same contract to different types of UEs.

From Lemma 5, we know that the BS will give different UEs the same (P, B) offer under specific types of utility/profit functions. The quantity is always equal to $(\frac{\beta}{\alpha c L})^{1/\alpha}$; and the price is determined by the type with lowest channel quality which the BS would like to serve with D2D service. From the sight of BS, a higher price leads to a higher per-UE profit, but it also implies fewer UEs are willing to choose D2D service, and vice versa. When we maximize the profit, we are looking for an optimal boundary channel quality the BS is willing to serve to balance between these two effects. We first provide the necessary condition under which the BS will give up a certain type of UEs.

Lemma 6. *Let $k \in \mathcal{N}$, if k satisfies the inequality*

$$\frac{a_k}{\sum_{i=k+1}^{N} a_i} \leq \frac{\alpha \ln(\frac{Q_{k+1}}{Q_k})}{\ln \frac{\beta Q_k^{\alpha}}{\alpha c L} - 1 - \frac{\beta}{\alpha} U_{cell}}, \tag{21}$$

forgoing D2D service profit from type 1 to k will be a possible solution for the BS pursuing the profit maximization.

When the inequality of Lemma 6 is satisfied, giving up from type 1 to type k potentially increases the revenue of BS. Nevertheless, the feasible k may not be unique. Remember that Lemma 6 only compares the difference of profit between serving type k or not. When the UEs can be categorized to many different types, there may exist a set of k satisfied the inequality. Therefore, in order to maximize the profit of the BS, we introduce the linear search algorithm to help the BS determining whether to serve these kinds of UEs with D2D mode.

Algorithm 4 A Linear Search Algorithm for UQBS System

Input: Number of UEs in each type: $a_i, i \in [1, N]$; Channel Quality of each type: $Q_i, i \in [1, N]$; Corresponding price of channel quality Q_i: P_i; UE's utility in cellular mode: U_{cell}; cost parameter: c; current load: L

Output: The most profitable choice K_{opt}.

```
Initial K_opt=1; profit_t = 0; profit*=(P_1 B* − cL × B*^α) Σ_{l=1}^{N} a_l.
Use the inequality in Lemma 6 to find all possible K_i, i ∈ [1, m], then store it.
for each j ∈ [1, m] do
    initial profit_t=P_{K_j} B* − cL × (B*)^α) Σ_{l=K_j}^{N} a_l.
    if profit_t > profit* then
        profit*= profit_t
        K_opt = K_j.
    end if
end for
Return K_opt.
```

The process of Algorithm 4 is to compute the optimal first to define a Possible Unavailable Set K, which denotes the set of UEs that the BS may not serve in D2D mode, and use it in the algorithm. With this step, we avoid the heavy load comparing the BS's profit of serving from i to n different types where $i \in [1, N]$ by limiting the search space to Possible Unavailable Set. Then, we use linear search to compare the BS's profit with each possible k to find the optimal solution.

With Algorithm 4, we know what types of UEs the BS should serve with D2D service. In practice, giving up such kinds of UEs can be achieved by setting a lower bound of channel quality in D2D mode. For those UEs with a channel quality below the lower bound, they can use cellular mode only; conversely, others with higher than lower-bound channel quality have the right to select a transmission mode between cellular and D2D as they like. Furthermore, the algorithm has linear complexity, which is much more efficient than exponential complexity in Algorithm 3. If the utility functions and profit functions in real world are close to the assumption of this case, uniform resource allocation with Algorithm 4 is a practical solution for the UCQ problem guaranteeing the efficiency of LTE-D2D systems.

VI. Numerical and Simulation Results

We would like to evaluate the performance of the proposed D2D-cellular systems and corresponding mechanisms. Specifically, we evaluate the utilities of UEs and also profit of the BS in the proposed system through numerical solutions and simulations under practical settings.

In general, all the notations in our numerical results and simulations are mentioned in Table I. The utility function and the profit function to get the numerical results are specified as follows. In this section, we set the benefit function $U_b(Q, B) = \beta\sqrt{(QB)}$ except for Fig. 4, payment function and the revenue function $U_c(B, P) = R_r(B, P) = B \times P$, the cost function $C(B, L) = cL \times B^2$, and the channel quality $Q = W \times log_2(1 + \text{SINR})$. Moreover, we set the same system parameters with satisfactory weighting factor $\beta = 10$, bandwidth per resource block $W = 180$kHz, and the product of cost weighting factor and current load $c \times L = 40$.

A. Welfare Comparison in UE-centric and BS-centric Protocol Design

As we discussed in Section refsec:PIUE, V, the difference between PIUE scenario and PIBS scenario is the control power of BS in D2D resource allocation. Here we compare the social welfare, utility, and profit in both PIUE and PIBS scenarios.

The social welfare of PIBS scenario shows that all benefits are collected by the BS as discussed in Section V. The social welfare in the PIUE scenario is the sum of profit of the BS and utility of the UE. The difference between PIBS and PIUE scenario is the cost of adopting a system with mechanism that the UE are willing to use D2D mode. The profit of the BS and utility of the UE show how the social welfare is distributed in PIUE scenario. When the channel quality becomes better, most of the additional welfare goes to the UE rather than the BS. This phenomenon can be ascribed to the UE's freedom of

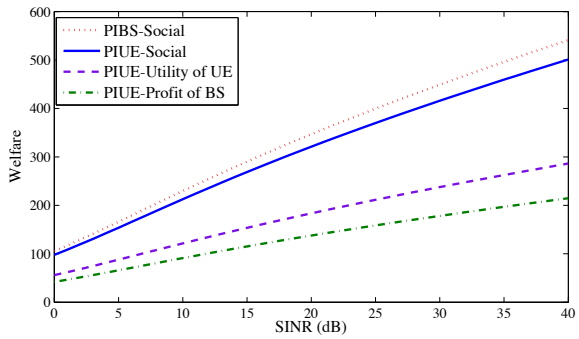

Fig. 2: Welfare in PIUE/PIBS Scenario

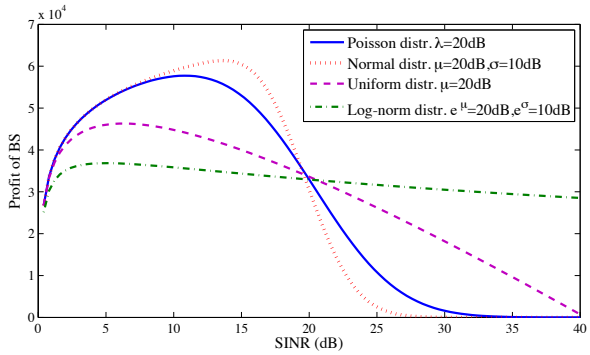

Fig. 3: Strategy versus profit of BS under different distributions

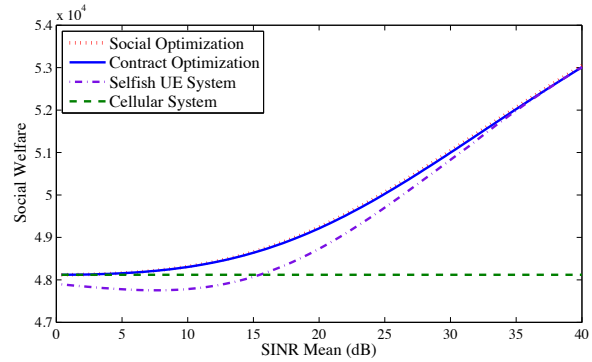

Fig. 4: Efficiency Comparison of Uniform Resource Allocation System

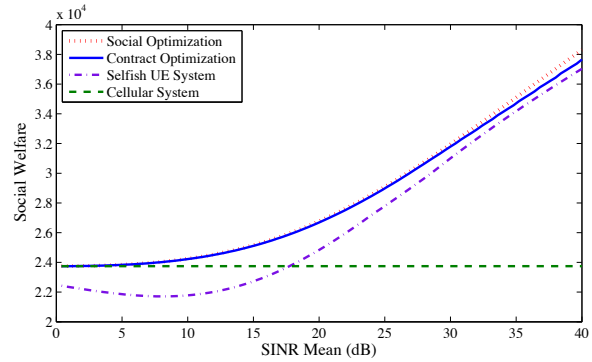

Fig. 5: Efficiency Comparison of Polynomial Benefit Function of UEs

B. Strategy of UQBS Varies with Distribution

In UQBS system, we would like to know how the BS's strategies will be affected under different distributions. In this subsection, we assume that there are 100 different types of UE with $\mathrm{SINR} = 0.4i\ \mathrm{dB}, i \in [1, 100]$. The BS's strategy is to set a lower bound of serving channel quality and serves all UEs with better channel qualities. We compare four different channel quality distributions to observe the impact of channel quality distributions in the BS's strategies.

Fig. 3 shows the strategy of BS in UQBS varies with distributions. This numerical analysis brings out two observations: first, if a system includes relatively more devices with higher channel quality, the BS will set a higher lower bound, and vice versa. Second, in a more concentrated distribution, the BS will set a lower bound closer to the average SINR. The phenomenon is reasonable because the less UEs in extremes, the less cost in BS's strategies of sacrificing the revenue from those types in extremes.

C. System Efficiency of UQBS under a 19-cell Simulator

In this simulation, we developed an LTE-D2D simulator following the 3GPP D2D specification [41], [42], [43] in order to analyze performance of proposed protocols in practical scenarios.

We create 19 macro cells in hexagon layout with total 75 pairs of UEs (potential D2D pairs) randomly and uniformly dropped within the clusters of the small circle area. The radius and the center of the circle area depends on the simulation scenario. We follow [44], [45] to implement the LTE D2D and of D2D communication comes from the UEs randomly located from neighboring cells. In this system, we set the UE's utility in cellular mode U_{cell} the same as utility at $\mathrm{SINR} = 7.82\ \mathrm{dB}$, which is derived from the SINR mean of UEs in the 19-cell system. We alter the average distance between D2D pairs to control the SINR mean in D2D mode. The SINR distributions of UEs are categorized into 100 types. Other detail parameters are shown in Table IV.

TABLE IV: Parameter settings

Parameter	Setting
Layout	Urban macro (500m ISD) + 1 Hotzone per macrocell
Per-resource Bandwidth	180kHz
Carrier Frequency	2GHz
Number of Cells	19
D2D Tx Power	23dBm
Macro Tx Power	46dBm

Finally, we demonstrate the efficiency improvement of the D2D cellular system in our solution and the difference between our solution to the social optimal system.

In Fig. 4, 5, the selfish UE mechanism in the figure represents that the BS is open to serve anyone who want to use D2D mode, so the UEs are free to choose D2D or cellular mode according to the channel quality experienced by the UE. The social optimization represents the social optimal system, which is a system that the BS knows the real channel quality of each UEs and can fully control the

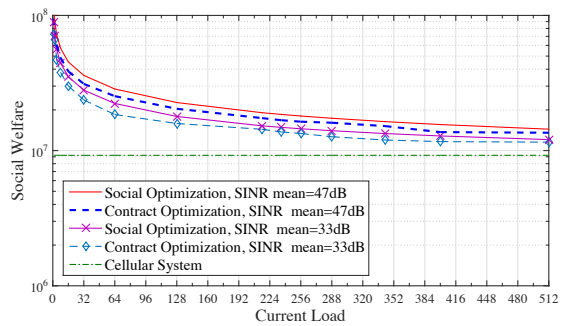

Fig. 6: Performance Degradation under Different Current Loads, $C(L, B) = L \times B^2$

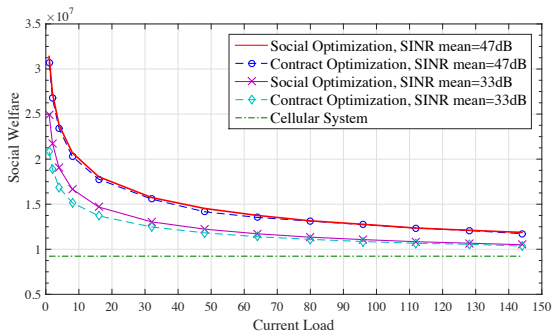

Fig. 7: Performance Degradation under Different Current Loads, $C(L, B) = L \times B^3$

Fig. 4 and Fig. 5 is the benefit function of UE. In Fig. 4, $U_b(Q, B) = 80\log(QB)$. It satisfies the condition of uniform resource allocation discussed in Section V-B with allocated resource blocks $B = (\frac{\beta}{\alpha c L})^{1/\alpha} = 1$. Hence, the Algorithm 4 can be used in this scenario. In Fig. 5, $U_b(Q, B) = 10\sqrt{QB}$, which is the same as other figures and Algorithm 3 is adopted.

Fig. 4, 5 show that our contract-based mechanism outperforms the selfish UEs mechanism in both scenario because our mechanism takes the loading and resource cost of the BS into consideration. Such mechanism rejects some UEs with poor channel quality to improve the system efficiency, and the performance is close to the social optimum.

D. System Efficiency under Different Current Loads

In this simulation, the system efficiency under different current loads has been provided. A 19-cell simulator the same as in Section VI-C is adopted and detail parameters are shown in Table IV. In order to know how the system efficiency degrades with the rising current load L, the mean of D2D-pair distances are fixed to control the SINR mean. The benefit function are defined as $U_b(Q, B) = 2048\sqrt{QB}$. The cost function is set as $C(L, B) = L \times B^2$ in Fig. 6 and $C(L, B) = L \times B^3$ in Fig. 7. Then, we average the performance under 10,000 different 19-cell systems to show that the performance degradation of different systems.

According to Fig. 6, 7, they show that the proposed contract-based mechanism outperforms the cellular system and performs close to the social optimum. It validates that the proposed contract-based mechanism can solve the UCQ problem under BS-centric protocol design, the system efficiency is close to social optimum. Such mechanisms rely on an assumption that the distribution of different type of UEs is known

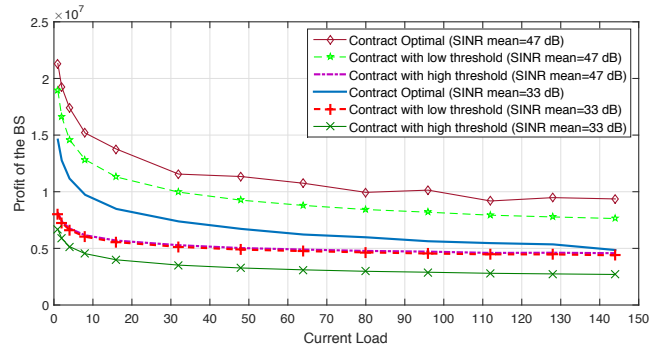

Fig. 8: Comparison of Profit of the BS under Different Serving Types

observe that the system efficiency of social optimum degrades with the increasing current loads. Therefore, in scenarios with high current load, the performance of social optimization and the contract-based mechanism will be close to the cellular systems because of the BSs rising cost of allocating resource.

E. Influence of Service Types

As discussed in Section V-B, an UE will truthfully report its channel quality if the contracts offered by the BS belong to the feasible contract sets, not necessarily belong to the optimal contract set of the optimal serving types. In this simulation, we want to demonstrate the importance of choosing proper serving types.

Giving all parameters the same as the simulation in Fig. 7, we compare the proposed algorithm with a naive threshold-based approach. The approach set a constant serving type threshold regardless of the loading and channel quality distributions, and then the BS will offer D2D service to any UE with its type above the pre-determined threshold. In Fig. 8, we compare the proposed algorithm with two thresholds, SINR= 0dB and SINR= 50dB. The proposed algorithm of choosing serving type gives the BS a higher profit than these two thresholds under different current load and different channel quality distributions. The simulation verifies that the choice of serving types is an important factor of the profit of the BS.

VII. Discussion

According to theoretical analysis and simulation results, we have demonstrated how the information state and the control ability of the BS affect the performance of the LTE-Advanced D2D system. In different information state (perfect information or UCQ state), the system is more efficient when the UCQ problem is not our concern. Moreover, the BS-centric protocol, which BS has more control ability (determining both D2D price and quantity rather than only D2D price), has a higher social welfare.

type of UEs requires the service operator to make extra efforts to collect and estimate before actually deployment. Generally, the more accurate estimation on the UE distribution, the more precise the contracts can be designed to extract the extra values from the UEs, and therefore more profits. However, even a rough one is also able to design the contracts with some efficiency loss due to imperfect service differentiation. Therefore, the service operator should find a balance between the accuracy of the UE distribution knowledge and the cost of extra efforts according to the profit in returns from the knowledge. The profit maximization design with imperfect estimation of UEs' channel quality distribution is an open problem and is considered as one of the possible future extensions of this work.

Additionally, the service operator can formulate different distribution models for different situations, and revise the model from some statistic data of the locations and behaviors of UEs based on the current observations. That is, the operator can change the D2D pricing model easily and can learn from the past experience to revise accordingly. Furthermore, based on the current observation and some prediction about upcoming load, a more general model taking the load L as an endogenous variable can be considered and some possible approaches, such as categorizing different usage of D2D users, may help the design and modeling the load L has been provided in [37]. Based on the updated UE distribution and up-to-date load L, we may consider a dynamic pricing model. This can also be considered as a direction of future extensions.

VIII. Conclusion

D2D communication has a great potential to be a widespread transmission approach under LTE-Advanced systems. In our paper, two different protocol designs of D2D resource allocation, UE-centric of BS-centric protocol designs are proposed. With the Stackelberg game approach, we modeled the selfish behaviors of the BS and UEs, and derived the equilibrium outcome of the systems. We focused on the Unknown Channel Quality (UCQ) problem which exists uniquely in D2D communication and has negative impact on the system performance. The defunct of the system in UE-centric protocol design due to the infinity iterations has been proved. In BS-centric system, we proposed a contract-based mechanism to maximize the profit of the BS and obviate the deviating of the UEs to resolve the issue of UCQ problem. Numerical and simulation results show that the proposed contract-based solution in BS-centric protocol design improves the system efficiency and performs closely to the optimal system.

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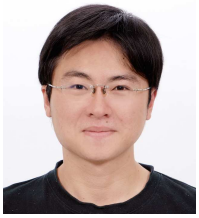

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APPENDIX

A. Proof of Lemma 1

Proof. Refer to Definition 4 in Section III-D, a set of strategy profile becomes an SPNE if all players cannot gain any profit from changing their own strategies in each subgame. In PIUE scenario, the UE's strategy is to determine the quantity of resource request in D2D service. We prove the lemma from left to right by contradiction. First, assume one of above inequalities is not satisfied.

For (9), $B^*_{D2D} < 0$ is trivial because we cannot allocate negative number of resource blocks. $B^*_{D2D} = 0$ means choosing cellular service.

For (10), if $U^*_{D2D}(Q, B^*_{D2D}, P^*_{D2D}) < U_{cell}$, it will not be an SPNE because the UE will switch to cellular mode for its interest.

Likewise, the the BS will switch to cellular mode for its interest if $R^*_{D2D}(B^*_{D2D}, P^*_{D2D}, L) < R_{cell}$.

That is to say, if any one of the inequality (9), (10), (11) is not satisfied, either the UE or the BS can choose to change their strategies into cellular service in order to get higher utilities.

On the other hand, we should prove the opposite direction. From inequality (10), (11), we can show that any feasible D2D service that satisfy the inequalities will outperform cellular service. Now all we need to do is to show that the equilibrium service (P^*_{D2D}, B^*_{D2D}) is an SPNE among all D2D strategies.

The service is combined by P_{D2D} and B_{D2D}, which is the strategy of BS and UE respectively.

$$\begin{aligned} & B^*_{D2D}(P') = \arg\max U_{D2D}(Q, B, P)|_{P=P'} \\ \Rightarrow\ & \forall P,\ U_{D2D}(B^*, P) \geq U_{D2D}(B, P),\ P^* \\ & = \arg\max R_{D2D}(B(P), P, L) \end{aligned} \tag{22}$$

Under the scenario that the BS determines its strategy first, its profit. At the same time, the UE's best strategy is B^*_{D2D}, the best response given any price. Both the UE and the BS cannot gain more utility or profit by changing their strategy, thus (P^*_{D2D}, B^*_{D2D}) is an SPNE among all D2D services set (P_{D2D}, B_{D2D}). With (9)-(11), we can further conclude that it outperforms the cellular service for both players. The proof is complete here. □

B. Proof of Lemma 2

Proof. Set $U_b(Q, B) = \sum_{i=0}^{n} \alpha_i (QB)^{\beta_i}$, $\forall \beta_i < 1$ (without loss of generality)

According to (6), the BS's objective function is as follows:

$$\begin{aligned} & \max_B \frac{\partial U_b(Q, B)}{\partial B} \times B - C(B, L) \\ = \quad & \max_B \sum_{i=0}^{n} \alpha_i \beta_i (QB)^{\beta_i} - C(B, L) \end{aligned} \tag{23}$$

Hence, B^*_{D2D} satisfies $\sum_{i=0}^{n} \alpha_i \beta_i \frac{\partial (QB)^{\beta_i}}{\partial B} - \frac{\partial C(B, L)}{\partial B} = 0$.

For any Q_1, Q_2, assume $Q_1 > Q_2$ without loss of generality, the following condition hold:

$$\sum_{i=0}^{n} \alpha_i \beta_i^2 Q_1^{\beta_i} (B^*_{D2D|Q_2})^{\beta_i - 1} > \left. \frac{\partial C(B, L)}{\partial B} \right|_{B = B^*_{D2D|Q_2}} \tag{24}$$

where $B^*_{D2D|Q_2}$ is the optimal amount of resource blocks under Q_2.

In equilibrium, the following equation should be satisfied:

$$\sum_{i=0}^{n} \alpha_i \beta_i^2 Q_1^{\beta_i} (B^*_{D2D|Q_1})^{\beta_i - 1} = \left. \frac{\partial C(B, L)}{\partial B} \right|_{B = B^*_{D2D|Q_1}} \tag{25}$$

Because $U_b(Q, B, P)$ is strictly concave increasing to QB and $C(B, P, L)$ is convex increasing to B, $B^*_{D2D|Q_1} > B^*_{D2D|Q_2}$ must holds to satisfy (24) (25). □

C. Proof of Lemma 4

Proof. Refer to (5), we know that the BS can always improve its profit by increasing the price P under any fixed quantity B. A rational UE will choose the D2D service only if $U_{D2D}(Q, B, P) \geq U_{cell}$. Assume the BS has maximized its profit with contract (P^*_{D2D}, B^*_{D2D}), if $U(Q, P^*_{D2D}, B^*_{D2D}) > U_{cell}$, let P'_{D2D} be the solution solved from $U(Q, P'_{D2D}, B^*_{D2D}) = U_{cell}$. Since the utility function $U(Q, P, B)$ strictly decreases in P and $U(Q, P^*_{D2D}, B^*_{D2D}) > U(Q, P'_{D2D}, B^*_{D2D})$, we can derive that $P'_{D2D} > P_{D2D}$.

Meanwhile, $R(P'_{D2D}, B^*_{D2D}, L) > R(P^*_{D2D}, B^*_{D2D}, L)$ because the profit of the BS increases in P. The result contradicts the assumption that the BS has already maximized its profit. □

D. Proof of Lemma 6

Proof. Initially, we serve all types of UEs with channel qualities from Q_1 to Q_N, the profit of the BS is $a_1 R_1 +$

$Q_1 \leq Q_2 \leq, \cdots, \leq Q_N$ without loss of generality, giving same contract for all UEs can simplify the function as:

$$\sum_{i=1}^{N} a_i(P_1B_1 - cLB_1{}^{\alpha})$$
$$where \qquad P_1B_1 = \frac{\beta}{\alpha}\ln(\frac{\beta Q_1^{\alpha}}{\alpha cL}) - U_{cell},\ cLB_1{}^{\alpha} = \frac{\beta}{\alpha} \tag{26}$$

If the BS chooses not to serve UEs from type 1 to type $k-1$, it will lose clients among these types, $\sum_{i=1}^{k-1} a_i$, but still serves type k to N. However, the lowest channel quality from serving type k to N is Q_k, which means the BS can set a new price P_k, treating type k as previous type 1, to increase its profit. The profit function of the BS is similar to (26), replacing all parameters from type 1 to type k.

When the BS decide whether to serve type k UEs or not, it checks if the following inequality holds:

$$\sum_{i=k+1}^{N} a_i(P_{k+1}B_{k+1} - cLB_{k+1}{}^{\alpha}) \geq \sum_{i=k}^{N} a_i(P_kB_k - cLB_k{}^{\alpha})$$

Then, we can substitute the result of Lemma 5 into the above equation and reduce it as follows.

$$\frac{a_k}{\sum_{i=k+1}^{N} a_i} \leq \frac{\alpha\ln(\frac{Q_{k+1}}{Q_k})}{\ln\frac{\beta Q_k^{\alpha}}{\alpha cL} - 1 - \frac{\beta}{\alpha}U_{cell}} \tag{27}$$

□

E. Proof of Theorem 1

Proof. A Pareto improvement improves the total welfare of a system. When communication in D2D mode is a Pareto improvement, it means the following inequality holds:

$$U_{D2D} + R_{D2D} > U_{cell} + R_{cell} \tag{28}$$

At first, let's assume that the BS did not aware the UE would cheat it. The honest strategy of UE gets the same utility as in Game-PIUE. Now, we want to know if lying to the BS with a lower channel quality Q is a better strategy for the UE.

For any Q_1, Q_2 with SPNE in D2D mode, there exist SPNE solutions (P_1^*, B_1^*), (B_2^*, B_2^*), respectively. Set $Q_2 = Q_1 - \epsilon$, $\epsilon \ll Q_2$ (without loss of generality), a fake channel quality report from Q_1 to Q_2 changes the UE's utility as follows:

$$\begin{aligned}
& U_{D2D}(Q_1, B_2^*, P_2^*) - U_{D2D}(Q_1, B_1^*, P_1^*) \\
= \ & U_b(Q_1, B_2^*) - U_b(Q_1, B_1^*) - (U_c(B_2^*, P_2^*) - U_c(B_1^*, P_1^*)) \\
= \ & \left.\frac{\partial U_b(Q,B)}{\partial B}\right|_{B=B_1^*} \times (B_2^* - B_1^*) - P_2^*B_2^* + P_1^*B_1^* \\
= \ & P_1^* \times (B_2^* - B_1^*) - P_2^*B_2^* + P_1^*B_1^* \\
= \ & (P_1^* - P_2^*) \times B_2^*
\end{aligned} \tag{29}$$

According to (29), if $P_1^* - P_2^* \neq 0$, the UE will lie to the BS for a higher utility. The only one situation that the UE will report the real channel quality Q when $P_1^* = P_2^*$. According to (6) and $P^* = \frac{\partial U_b(Q,B)}{\partial B}\big|_{B=B^*}$, the truthful-report condition exists only when $\frac{\partial^2 U_b(Q,B)}{\partial B \partial Q} = 0$.

However, $\frac{\partial^2 U_b(Q,B)}{\partial B \partial Q} = 0$ violates the assumption in III-E-2a

F. Proof of Theorem 2

Proof. Without loss of generality we assume $Q_1 \leq Q_2 \cdots \leq Q_N$, and denote UEs with channel quality Q_i as UE^i. Since contract 1 (B_1, P_1) for UE^1 can be dealt as in PIBS scenario, here we put emphasis on the analysis of contracts (B_i, P_i) and (B_{i-1}, P_{i-1}), where $i \in [2, N]$.

Given the feasible contract (B_{i-1}, P_{i-1}) under which UE^k, $k \in [1, i-1]$ reveal their real channel qualities, contract (B_i, P_i) has to be chosen carefully to avoid UE^{i-1} reporting a higher channel quality and avoid UE^i from reporting a lower one for more utility. We can conclude that the feasible contract (B_i, P_i) has only two possible results:

$$B_i = B_{i-1}, P_i = P_{i-1} \text{ or } B_i > B_{i-1}, P_i > P_{i-1}.$$

The case with the decrease of both B and P is not feasible due to the property of increasing marginal benefit $\frac{\partial U_b}{\partial B}$. Since the decrease in payment U_c for UE^{i-1} is as same as UE^i, which is equal to the increase in benefit U_b for UE^i. Thus UE^{i-1} can get greater utility choosing contract (B_i, P_i) than (B_{i-1}, P_{i-1}). We may follow similar procedures to check other cases. Note that in some utility functions and classification of qualities, it's impossible to find a different contract to distinguish different qualities. Thus the two types will share the same contract.

To prove the feasible contract set guarantees the honest report, we first prove that the type UE^i will never choose any contract (B_j, P_j) when $i > j$.

Assume $1 \leq j < i-1$ without loss of generality

$$\begin{aligned}
& B_{i-1} \geq B_j \\
\Rightarrow \ & U_b(Q_i, B_{i-1}) - U_b(Q_{i-1}, B_{i-1}) \\
& \geq U_b(Q_i, B_j) - U_b(Q_{i-1}, B_j) && (30) \\
\Longleftrightarrow \ & U(Q_i, B_{i-1}, P_{i-1}) - U(Q_{i-1}, B_{i-1}, P_{i-1}) \\
& \geq U(Q_i, B_j, P_j) - U(Q_{i-1}, B_j, P_j) && (31) \\
\Rightarrow \ & U(Q_i, B_{i-1}, P_{i-1}) - U(Q_i, B_j, P_j) \\
& > U(Q_{i-1}, B_{i-1}, P_{i-1}) - U(Q_j, B_j, P_j) && (32) \\
= \ & U(Q_i, B_{i-1}, P_{i-1}) - U(Q_i, B_j, P_j) > 0 && (33)
\end{aligned}$$

By definition, $U(Q_i, B_i, P_i) \geq U(Q_i, B_{i-1}, P_{i-1})$. Hence, UE^i will never choose any contract (B_j, P_j) when $i > j$. Likewise, let $i+1 < j \leq N$ without loss of generality:

$$\begin{aligned}
& B_{j-1} \leq B_j \\
\Rightarrow \ & U_b(Q_i, B_{j-1}) - U_b(Q_{j-1}, B_{j-1}) \\
& \geq U_b(Q_i, B_j) - U_b(Q_{j-1}, B_j) && (34) \\
\Longleftrightarrow \ & U(Q_i, B_{j-1}, P_{j-1}) - U(Q_{j-1}, B_{j-1}, P_{j-1}) \\
& \geq U(Q_i, B_j, P_j) - U(Q_{j-1}, B_j, P_j) && (35) \\
\Rightarrow \ & U(Q_i, B_{j-1}, P_{j-1}) - U(Q_i, B_j, P_j) \\
& \geq U(Q_{j-1}, B_{j-1}, P_{j-1}) - U(Q_{j-1}, B_j, P_j) && (36) \\
= \ & U(Q_i, B_j, P_j) - U(Q_i, B_{j-1}, P_{j-1}) < 0 && (37)
\end{aligned}$$

We can conduct that UE^i will never choose any contract (B_j, P_j) when $i < j$. From (33), (37), feasible contract set guarantees that all UEs will choose the contracts designed for their own types. Therefore, feasible contract set assures that all UEs report channel quality honestly.

□